CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 1 Analysis & Sorting

Practice on Lecture 1 and Lecture 2 · 11 questions

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01 Before you start

These questions are my own, built on the examples of the two lectures and on the style of the past midterms. The solutions are in a separate file.

- Q1 to Q3: counting primitive operations, a fixed midterm question.
- Q4 to Q7: Big-Oh and concepts.
- Q8 to Q11: prefix averages and sorting.

The counting rule

Each assignment, array index, comparison, arithmetic operation or return counts as one operation. A `for i ← a to b` line costs 1 for the initialization plus the number of tests (the number of iterations plus 1).

02 Questions

Q1 COUNTING OPERATIONS

Count the total primitive operations of the following algorithm, then give its running time in Big-Oh.

```
Algorithm arraySum(A, n)
    sum ← 0
    for i ← 0 to n − 1 do
        sum ← sum + A[i]
    { increment counter i }
    return sum
```

Q2 COUNTING OPERATIONS · MIDTERM STYLE

Count the total primitive operations of the following algorithm:

a) in the worst case. b) in the best case.

```
Algorithm arrayMax(A, n)
   currentMax ← A[0]
   Temp ← A[0]
   for i ← 1 to n − 1 do
      if A[i] > currentMax then
         currentMax ← A[i]
   { increment counter i }
   return currentMax
```

Q3 COUNTING OPERATIONS · NESTED LOOPS

Count the total primitive operations of the following algorithm, then give its running time in Big-Oh.

```
Algorithm countPairs(n)
    count ← 0
    for i ← 0 to n − 1 do
        for j ← 0 to n − 1 do
            count ← count + 1
        { increment counter j }
    { increment counter i }
    return count
```

Q4 BIG-OH DEFINITION

Using the definition of Big-Oh, find constants c and n_0 that show:

a) $5n + 3$ is $O(n)$

b) $4n^2 + 3n + 2$ is $O(n^2)$

c) $3 \log n + 5$ is $O(\log n)$

d) Show that n^3 is not $O(n^2)$.

Q5 GROWTH RATES

Order the following functions from the slowest growth rate to the fastest:

2^n , $5n$, $n \log n$, 100 , n^3 , $\log n$, $4n^2$

Q6 BIG-OH RULES

Write each function in the simplest Big-Oh form:

a) $10^5n^2 + 10^8n$

b) $n(n + 1) / 2$

c) $3 \log n + \log \log n$

d) $7n - 2$

e) $2^n + n^3$

f) 500

Q7 TRUE / FALSE

Put T or F for each of the following statements:

1. In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size.
2. Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.
3. The total of primitive operations represents the worst case of an algorithm.
4. Changing the hardware environment changes the growth rate of T(n).
5. The function n^2 is O(n).
6. Merge sort runs in O(n log n) time only in the best case.
7. The best case of insertion sort takes O(n) steps.
8. Quick sort requires an auxiliary array and a merge step.
9. In the RAM model, accessing any cell in memory takes unit time.
10. f(n) is Ω(g(n)) if f(n) is asymptotically less than or equal to g(n).

Q8 PREFIX AVERAGES

Let X = [4, 8, 6, 2].

a) Compute the array A of prefix averages of X.

b) Why does prefixAverages1 run in $O(n^2)$ time?

c) Write an algorithm that computes the prefix averages in O(n) time.

Q9 INSERTION SORT

a) Apply insertion sort to the array [5, 2, 4, 6, 1, 3]. Show the array after each iteration of the outer loop.

b) What input gives the best case and what input gives the worst case? Give the running time of each.

c) State the three steps of a loop-invariant proof.

Q10 MERGE SORT

a) Apply merge sort to the array [7, 2, 9, 4, 3, 8, 6, 1]. Show the divide steps and the merge steps.

b) Write the equation of T(n), the time to run MERGESORT on a list of size n.

c) For n = 16, in the recursion tree: how many levels are there? How many problems are at level 3 and what is the size of each? What is the total work if each level costs 6n?

Q11 DIVIDE AND CONQUER

a) What are the three steps of the divide-and-conquer strategy?

b) Give three differences between merge sort and quick sort.

c) What is the worst-case running time of each?